NCERT · CLASS 7 · MATHS · PART II

# Operations with Integers

C02

## Positive, negative and zero

**Number lines · Tokens · Sign rules**

**Six NCERT exercise sets with worked steps**

Textbook answers first · Visual learning · Quick revision

Original editable teaching notes based on NCERT; illustrations are schematic.

# Part A · NCERT Figure it Out / Try This answers

**Using these answers**

The NCERT Part II chapter is printed as Chapter 2 (pp. 24–46). Questions are grouped by textbook page and number. Work through the sign rule before checking each answer. A few open-ended tasks have more than one correct solution.

## Set 1 · Sums, differences and carrom coin (pp. 24–27)

### Rakesh's puzzle · six integer pairs

If first number is $x$ and second $y$, then $x+y=S$, $x-y=D$, so $x=(S+D)/2$ and $y=(S-D)/2$.

| Part | Sum | Difference | Pair $(x,y)$ |
|---|---|---|---|
| (a) | 27 | 9 | $(18,9)$ |
| (b) | 4 | 12 | $(8,-4)$ |
| (c) | 0 | 10 | $(5,-5)$ |
| (d) | 0 | $-10$ | $(-5,5)$ |
| (e) | $-7$ | $-1$ | $(-4,-3)$ |
| (f) | $-7$ | $-13$ | $(-10,3)$ |

### Carrom-coin movements

Starting at 0, movements add with signs. $5+(-7)=\boxed{-2}$; the coin finishes 2 units left of zero.
(i) If the first movement is $-4$ and the final position 5, then $-4+b=5$, so $\boxed{b=9}$.
(ii) The series $1,-2,3,-4,\ldots,9,-10$ is five pairs each summing $-1$, so the final position is $\boxed{-5}$.
In the diagram tasks, compare the total rightward and leftward lengths of arrows $a,b$; the sign of the endpoint $P$ tells which total is larger.

### Token model · subtraction

A green $+1$ and red $-1$ make one **zero pair**. Hence $7-18=7+(-18)=\boxed{-11}$, and $4-(-12)=4+12=\boxed{16}$. Adding zero pairs changes no total, but lets us remove the required tokens.

## Set 2 · Multiplication tokens (p. 31)

### Q1 · Find the four products

(a) $3(-2)=\boxed{-6}$; (b) $(-5)(-2)=\boxed{10}$; (c) $(-4)(-1)=\boxed{4}$; (d) $(-7)3=\boxed{-21}$.
Same signs $\Rightarrow$ positive; different signs $\Rightarrow$ negative.

### Q2 · Use $123\times456=56088$

(a) $(-123)456=\boxed{-56088}$; (b) $(-123)(-456)=\boxed{56088}$; (c) $123(-456)=\boxed{-56088}$. Magnitudes stay the same; only the sign rule changes.

### Q3 · The multiplication rule

First multiply the positive magnitudes. Put + if both numbers have the same sign; put − if they have different signs. A factor zero makes the product zero.
For any token arrangement representing −2, four copies still represent $4(-2) = \boxed{-8}$. For arrangements representing +4, five copies give $\boxed{20}$; adding zero pairs does not change the total.

## Set 3 · Pattern products (pp. 33–34)

### Q1 · Six products

(a) $4(-3) = \boxed{-12}$; (b) $(-6)(-3) = \boxed{18}$; (c) $(-5)(-1) = \boxed{5}$; (d) $(-8)4 = \boxed{-32}$; (e) $(-9)10 = \boxed{-90}$; (f) $10(-17) = \boxed{-170}$.

### Patterns and swapping factors

When the first factor decreases by 1, the product changes by the negative of the second factor. Example $4(-3) = -12$, $3(-3) = -9$, $2(-3) = -6$: each step rises by 3.
$(-30)12 = 12(-30) = \boxed{-360}$ and $14(-5) = (-5)\boxed{14} = -70$. For all integers, $ab = ba$ (commutative law).

## Set 4 · Division and real-life integers (p. 39)

### Q1 · Multiplication and division

(a) $14(-15) = \boxed{-210}$; (b) $(-16)(-5) = \boxed{80}$; (c) $36 \div (-18) = \boxed{-2}$; (d) $(-46) \div (-23) = \boxed{2}$.

### Q2 · Freezing a room

Start 32°C, lower by 5°C each hour for 10 hours: $32 + 10(-5) = 32 - 50 = \boxed{-18^\circ\text{C}}$.

### Q3 · Cement profit and loss

(a) White: $3000(8) = 24000$. Grey: $5000(-5) = -25000$. Net $24000 - 25000 = \boxed{-1000}$: a loss of 1000 rupees.
(b) At 6400 grey bags the loss is $6400 \times 5 = 32000$. White bags needed $= 32000/8 = \boxed{4000}$.

### Q4 · Fill the six blanks

(a) −9; (b) −7; (c) 7; (d) −11; (e) −56; (f) −132.
Check: $(-3)(-9) = 27$, $7(-8) = -56$, and $(-56)/(-8) = 7$.

## Set 5 · Expressions, properties and patterns

### Multiplication and distributive property

Multiplication is associative: $(5 \cdot (-3)) \cdot 4 = 5 \cdot ((-3) \cdot 4) = \boxed{-60}$. The product $25(-6)12 = \boxed{-1800}$ whichever order you use.
Distribute before simplifying: $(-2)[4+(-3)] = (-2)(1) = -2$, and $(-2)4+(-2)(-3) = -8+6 = \boxed{-2}$.
For $(-1)$ multiplied $n$ times: **even** $n \Rightarrow +1$, **odd** $n \Rightarrow -1$. An explicit sign typo in the textbook's five-factor printed line does not change this rule.

### Pattern machines · textbook p. 41–43

**Machine 1:** $a + b - c$ (given in the chapter). Its example $(-10) + (-12) - (-9) = \boxed{-13}$.
**Machine 2 / Figure it Out Q7:** The rule is $a - (b \times c)$: subtract the product of the second and third inputs from the first.
Check: $4 - 8(-3) = 28$, $6 - 9(6) = -48$, $2 - 3(-2) = 8$, $-9 - 5(-8) = 31$, and $7 - (-4)(-6) = -17$.
For inputs $-16, -6, -9$, output $= -16 - [(-6)(-9)] = -16 - 54 = \boxed{-70}$.

## Set 6 · Final Figure it Out (pp. 42–44)

### Q1 · Evaluate products

(a) $(-5)(18 - 3) = (-5)15 = \boxed{-75}$.
(b) $(-7)4(-1) = \boxed{28}$.
(c) $(-2)(-1)(-5)(-3) = \boxed{30}$ (four negative factors).

### Q2 · Evaluate divisions

(a) $-27/9 = \boxed{-3}$; (b) $84/(-4) = \boxed{-21}$; (c) $(-56)/(-2) = \boxed{28}$.

### Q3–Q4 · Additive inverses

Q3: To get $27, -31, -1, 1, 0$ by multiplying by $-1$, the required integers are, respectively, $\boxed{-27, 31, 1, -1, 0}$.
Q4: Reversing every sign in the given expression $-4$ changes the result to $\boxed{4}$.

### Q5 · Modified Collatz exploration

The rule is: even $n \mapsto n/2$; odd $n \mapsto -3n + 1$.
For $-6$: $-6 \to -3 \to 10 \to 5 \to -14 \to -7 \to 22 \to 11 \to -32 \to -16 \to -8 \to -4 \to -2 \to -1 \to 4 \to 2 \to 1 \to -2$ (repeat).
For $-21$: $-21 \to 64 \to 32 \to 16 \to 8 \to 4 \to 2 \to 1 \to -2 \to -1 \to 4$ (repeat).
These examples eventually enter a repeating cycle; this alone does not prove all starting integers do.

### Q6 · Marking test

(a) 15 correct answers earn $15 \times 4 = 60$. To score 40, wrong answers cost 20, so $20/2 = \boxed{10}$ wrong and $\boxed{25}$ total questions.
(b) Five correct give 20 points; to score $-10$, the penalty is 30, so $\boxed{15}$ wrong. The same test has 25 questions from part (a), so $25 - (5 + 15) = \boxed{5}$ unanswered.

### Q8–Q9 · Temperature and consecutive numbers

Q8: $8 + 4(-5) = \boxed{-12^\circ\text{C}}$.
Q9(a): $(-3)(-2)(-1) = \boxed{-6}$. Q9(b): $4 \cdot 5 \cdot 6 = \boxed{120}$.

## Q10 · Coins worth $+13$ and $-9$ pibs

If $x, y$ are numbers of $+13$ and $-9$ coins, the total is $13x - 9y$. Examples with $x, y \geq 0$:

| Target | 20 | 40 | $-50$ | 8 | 10 | $-2$ | 1 | 1568 |
|---|---|---|---|---|---|---|---|---|
| $x$ | 5 | 10 | 1 | 2 | 7 | 4 | 7 | 122 |
| $y$ | 5 | 10 | 7 | 2 | 9 | 6 | 10 | 2 |

**Yes, 1568 is possible:** $122(13) - 2(9) = 1586 - 18 = 1568$. These are possible combinations, not the only ones.

## Q11 · Fractions from integer operations

Respect parentheses, multiplication and division grouping.

(a) $\dfrac{32(-18)}{-36} = \boxed{16}$. (b) $\dfrac{32}{(-36)(-18)} = \boxed{\frac{4}{81}}$.

(c) $\dfrac{25(-12)}{45(-27)} = \boxed{\frac{20}{81}}$. (d) $\dfrac{280(-7)}{(-8)(-35)} = \boxed{-7}$.

## Q12 · Increasing order

Values: (a) $-1412$, (b) 716, (c) 1412, (d) 370272, (e) $-370272$, (f) 335472.

Thus $\boxed{(e) < (a) < (b) < (c) < (f) < (d)}$.

## Q13–Q14 · Calculate by reasoning

Given $(-548)972 = -532656$:

(a) $(-547)972 = -532656 + 972 = \boxed{-531684}$.

(b) $(-548)971 = -532656 + 548 = \boxed{-532108}$.

(c) $(-547)971 = \boxed{-531137}$.

Q14: $-207(33 - 7) = \boxed{-5382}$ (same value as $207(-33 + 7)$).

## Q15 · Use 3, $-2$, 5, $-6$ once

Use the three operations $+, -, \times$ once each (with brackets):

Maximum: $(-6)[-2 - (3 + 5)] = (-6)(-10) = \boxed{60}$.

Minimum: $(-6)[(3 + 5) - (-2)] = (-6)(10) = \boxed{-60}$.

These extrema were checked across the possible binary bracketing arrangements and number/operation permutations.

## Q16 · Five ways for each blank pattern

**(a)** $a + b \times c = -36$:

$0 + (-6) \times 6$, $4 + (-5) \times 8$, $-36 + 0 \times 7$, $-12 + (-4) \times 6$, $-40 + 2 \times 2$.

**(b)** $(a - b) \times c = 12$:

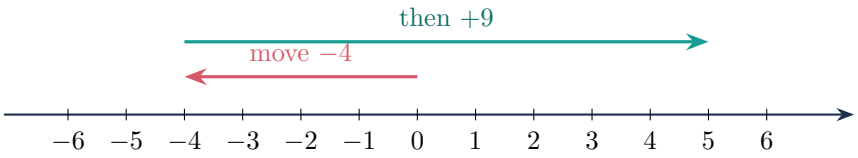
$(5 - 1) \times 3$, $(1 - 5) \times (-3)$, $(8 - 2) \times 2$, $(0 - (-6)) \times 2$, $(15 - 3) \times 1$.

**(c)** $a - (b - c) = -1$:

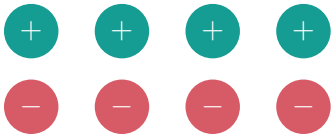
$0 - (2 - 1)$, $2 - (4 - 1)$, $-1 - (3 - 3)$, $3 - (5 - 1)$, $-4 - (0 - 3)$.

## Part B · Learn integer operations visually

**1. Integer number line.** Right is positive; left is negative.




**2. Zero pairs.** One positive and one negative token cancel each other.



Four zero pairs = 0

**3. The complete sign rules.**

| Signs | Multiply | Divide (nonzero divisor) |
|---|---|---|
| $+,+$ | $+$ | $+$ |
| $+,-$ | $-$ | $-$ |
| $-,+$ | $-$ | $-$ |
| $-,-$ | $+$ | $+$ |

**4. A multiplication pattern.** Observe the change in products:

$$4(-3) = -12, \quad 3(-3) = -9, \quad 2(-3) = -6, \quad 1(-3) = -3, \quad 0(-3) = 0, \quad (-1)(-3) = 3.$$

**5. Properties that help simplify.**

| Name | Rule and example |
|---|---|
| Commutative | $ab = ba$: $(-2)5 = 5(-2)$ |
| Associative | $a(bc) = (ab)c$: $2[(-3)4] = [2(-3)]4$ |
| Distributive | $a(b + c) = ab + ac$: $(-2)(4 - 3) = -8 + 6$ |

## Common mistakes · fix them fast

- $-4 - (-7) = 3$, *not* $-11$: subtracting a negative means adding its inverse.
- $(-4)(-7) = 28$, *not* $-28$: two negative signs multiply to positive.
- $0 \div 7 = 0$, but **division by zero is not defined**.
- $32 \div [(-36)(-18)]$ means divide by the *whole product*; do not compute from left to right across the brackets.
- If a story says "−2 marks for wrong answers", represent each wrong answer as $-2$ rather than adding $+2$.

**Three-minute practice**

1. $(-8)(-6) =$ 2. $-45 \div 9 =$ 3. $12 - (-9) =$
4. Start at $-3$ and move $+11$: endpoint?
5. $(-2)[4 + (-5)] =$

**Answers:** 1. 48; 2. $-5$; 3. 21; 4. 8; 5. 2.